Axient: On-Chain Credit and Loss Allocation for Leveraged Event Markets:

A Venue-Agnostic Protocol for Traders, Credit Providers, Market Makers, and Liquidation Backstops

Maksym Nechepurenko*

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Abstract

A physically backed leveraged event position requires real credit: when a trader contributes collateral C and receives leverage L, the protocol supplies $(L-1)C$ in stablecoin and uses the combined amount to acquire recognized event exposure. Once third-party capital finances that leg, the product becomes a credit protocol whose design must specify lender priority, liquidation authority, withdrawal liquidity, and loss allocation. This paper develops a venue-agnostic on-chain architecture for that layer. It separates traders, Senior Credit LPs, market makers, liquidators, and Liquidation Backstop Providers (LBPs), which supply junior loss-absorbing capital ahead of Senior principal. The model formalizes debt and pool shares, utilization- and risk-sensitive rates, collateral-locked position accounts, venue capability vectors, bonded market-maker commitments, isolated pools, non-redeemable reserves, loss-participating withdrawal queues, and a deterministic waterfall. Formal results establish settlement-confirmed debt priority, trader ownership of residual value, replay-safe partial settlement, non-dilutive share issuance with a maintained zero-mint guard, integer accounting closure, uniqueness and conservation of the waterfall, junior-before-Senior impairment, no reuse of shared protection capital, scenario-conditional Senior protection under frozen mandate-eligible junior loss-absorbing capital, request-time neutrality of queued shares, and a rationed utilization equilibrium with explicit clipping. A fixed-seed agent model evaluates 96 two-year paths with 636 heterogeneous agents per path and records 70,207,488 scalar invariant evaluations with zero failures. Layered protection reduces but does not eliminate Senior loss; 5×-heavy leverage materially increases capital risk; LBP capital is repeatedly impaired; and modest

*Founder and Director of Research, ForesightFlow, the Research Department of Devnull FZCO, Dubai, United Arab Emirates. Email: maksym@devnull.ae.

bonded market-maker capacity can expand admission faster than it reduces unit shortfall. These are synthetic mechanism comparisons, not forecasts of yield, default probability, venue liquidity, or production safety.

Keywords: event markets; on-chain credit; Senior liquidity providers; liquidation backstops; market makers; withdrawal queues; loss allocation; agent-based simulation.
JEL Classification: G13, G14, G21, G23, G28.

1 Introduction

A leveraged event position is not created by writing a larger number in a user interface. If a trader posts C and receives leverage L, gross exposure and debt are

$$N = LC, \qquad D_0 = (L-1)C. \tag{1}$$

For a physically backed implementation, the full amount N purchases recognized outcome exposure. The borrowed portion must therefore come from a treasury, credit pool, or partner balance sheet. This transforms a leveraged trading feature into a capital market connecting demand for event exposure with supply of credit and execution liquidity.

The closest DeFi analogues are overcollateralized lending protocols, but the risk geometry is different. A conventional lending borrower posts collateral exceeding debt; an event-margin trader may begin with only $1/L$ of gross exposure as equity, while the financed outcome claim can lose value rapidly or become non-tradable near close. Consequently, protection depends not only on collateral value but on enforceable control, early deleveraging, settlement evidence, shared-book capacity, reserves, and junior capital. The protocol is closer to on-chain secured margin finance than to ordinary overcollateralized money markets [Leshner and Hayes, 2019, Aave, 2026, Werner et al., 2022].

This paper builds the protocol-level complement to the position-level debt-free-finality mechanism. The concise revision retains the core accounting, theorem statements, endogenous-capital model, and principal evidence. Extended proofs, parameter registries, and diagnostic panels remain available in the earlier arXiv version history.

Contributions.

(C1) A four-sided protocol model for traders, Senior LPs, market makers, and LBPs, with liquidators treated as an execution role rather than a capital class.

(C2) On-chain balance-sheet accounting that separates accrued receivables from settled cash and reserves from redeemable provider shares.

(C3) A deterministic debt-first residual rule, replay-safe partial settlement, integer rounding policy, and explicit dust account.

(C4) A unique ordered loss waterfall and loss-participating withdrawal queue.
(C5) Venue-capability and market-maker commitment layers that remain explicit about external execution limits.
(C6) An endogenous capital model and fixed-seed agent experiment that exposes leverage crowding, run dynamics, wrong-way pricing, and common-factor contagion.

2 Relation to lending, financial networks, and agent models

The design draws on several adjacent literatures but combines them in a distinct way. DeFi lending protocols establish on-chain share accounting, utilization-based rates, collateral factors, and liquidation incentives [Leshner and Hayes, 2019, Aave, 2026, Werner et al., 2022]. Those systems generally begin from overcollateralized debt. Event-margin credit instead finances an asset whose executable value can deteriorate sharply near a known or endogenous close, so execution capacity and settlement evidence become first-class credit variables.

Financial-network models show how local balance sheets can transmit losses through common exposures, shared liquidity, and cross-guarantees [Eisenberg and Noe, 2001, Gai and Kapadia, 2010, Acemoglu et al., 2015]. Axient uses isolated local ledgers to prevent direct liability transfer, but it separately models protection contagion through shared reserves and common-factor covariance through stablecoins, venues, or market makers.

Agent-based finance provides a natural method for studying feedback among heterogeneous demand, capital supply, inventory, and price formation [Tesfatsion, 2006, Farmer and Foley, 2009, Brock and Hommes, 1998, Lux and Marchesi, 1999]. The present agent model is not offered as a calibrated market replica. Its purpose is mechanism comparison: hold one synthetic environment fixed, change the protocol layers, and record which qualitative effects survive an independent implementation.

The closest conceptual distinction is between a protocol claim and a company profit share. Senior and LBP participants receive contractual on-chain claims defined by the waterfall; they do not automatically receive equity-like participation in Axient's corporate profit. This separation makes the capital contract auditable and avoids disguising provider risk as a marketing return.

3 Roles, venue capabilities, and pool mandates

Definition 3.1 (Economic roles). *A* Trader *posts collateral and requests financed event exposure. A* Senior Credit LP *supplies stablecoin and receives the highest provider-priority claim. A* Liquidation Backstop Provider *supplies junior loss-absorbing provider capital* *ahead of Senior principal. A* Liquidator *executes an admissible risk-reducing transition for a bounty. A* Market Maker *supplies executable liquidity, RFQ capacity, or a bonded close commitment.*

One organization may perform several roles, but the ledger must keep them distinct. In particular, a liquidator action and an LBP balance sheet are not the same economic object.

A venue adapter publishes a capability vector rather than a binary "integrated" flag. Relevant coordinates include custody location, contract-account support, signer delegation, withdrawal lock, lender lien, liquidation authority, settlement proof, close observability, redemption proof, and permissionless execution. A pool mandate determines which capability classes, markets, leverage tiers, concentration limits, and backing modes its capital may finance.

The main backing modes are:

(a) contract-controlled physical backing;
(b) verifiable on-chain venue settlement with operator-gated matching;
(c) attested or custodial backing; and
(d) protocol-native synthetic exposure with external hedging.

They are not risk-equivalent. Senior capital should not silently move across them.

4 On-chain accounting

4.1 Position and pool balance sheets

Let $X_{i,t}$ be settlement-recognized value controlled by position i and $D_{i,t}$ its debt. Trader equity and latent credit shortfall are

$$E_{i,t} = [X_{i,t} - D_{i,t}]^+ , \qquad \ell_{i,t} = [D_{i,t} - X_{i,t}]^+ , \tag{2}$$

so

$$X_{i,t} + \ell_{i,t} = D_{i,t} + E_{i,t}. \tag{3}$$

Trader equity is first loss. Matched but unsettled proceeds are excluded from $X_{i,t}$.

For pool p, recognized assets are cash $A_{p,t}$ plus debt receivables $B_{p,t}$:

$$\mathcal{A}_{p,t} = A_{p,t} + B_{p,t}. \tag{4}$$

Claims consist of Senior NAV V^S, LBP NAV V^J, market and pool reserves, and retained protocol claims. Book closure requires equality of recognized assets and claims, but does not imply that every receivable is liquid or riskless.

4.2 Debt-first residual ownership

Let settlement-confirmed cash y enter a position with debt D_i and pre-authorized senior fees F_i^{due}. Allocation is

$$y_i^D = \min\{y, D_i\}, \tag{5}$$
$$y_i^F = \min\{\left[y - y_i^D\right]^+, F_i^{\mathrm{due}}\}, \tag{6}$$
$$y_i^T = \left[y - y_i^D - y_i^F\right]^+. \tag{7}$$

Proposition 4.1 (Trader residual ownership)**.** *Any settlement-confirmed value remaining after recognized debt and admitted senior fees belongs to the trader position claim. It cannot be credited to a reserve, provider class, or protocol treasury without a separate authorized transfer.*

4.3 Debt shares and settlement evidence

With borrow index $I_{p,t}^B > 0$ and debt shares d_i,

$$D_{i,t} = d_i I_{p,t}^B, \qquad I_{p,t+1}^B = I_{p,t}^B(1 + r_{p,t}^B \Delta t). \tag{8}$$

A new borrow mints debt shares; only a settlement-confirmed repayment transition can burn them.

A match j has capacity M_j. Settlement may arrive under unique evidence identifiers e with amounts $s_{j,e}$:

$$S_j = \sum_{e \in \mathcal{E}_j} s_{j,e} \le M_j. \tag{9}$$

The evidence identifier is consumed atomically with allocation.

Proposition 4.2 (Settlement-confirmed priority and replay safety)**.** *Matched, pending, reverted, or merely attested proceeds cannot reduce debt. If each evidence identifier is unique and cumulative settlement is capped by matched capacity, replay cannot reduce debt, increase provider cash, or release trader residual twice.*

4.4 Provider shares, reserves, and integer arithmetic

For Senior share supply Z^S and Senior NAV V^S, share price is $\vartheta^S = V^S/Z^S$. Deposit a mints a/ϑ^S shares, leaving incumbent price unchanged. LBP shares are analogous but carry junior impairment.

Definition 4.3 (Junior loss-absorbing capital and withdrawal headroom)**.** *Let $V_{p,t}^J$ be recognized junior-provider NAV. The quantity $J_{p,t}^{\mathrm{abs}}$ is the portion of that NAV which is mandate-eligible for impairment, reserved to an active protection certificate, and unavailable for service or reuse until the certificate is released or the capital is impaired. The* *quantity $J^{\text{free}}_{p,t}$ is the disjoint withdrawal-service headroom remaining after encumbrances, active protection reservations, required buffers, and policy haircuts. Thus*

$$0 \le J^{\text{abs}}_{p,t},\ J^{\text{free}}_{p,t}, \qquad J^{\text{abs}}_{p,t} + J^{\text{free}}_{p,t} \le V^{J}_{p,t}. \tag{10}$$

There is no general equality between the two quantities. A queued but unpaid share remains outstanding and loss-participating; it may be included in J^{abs} only while service of the corresponding value is blocked by the active certificate. Once payment and burn occur, that value is excluded from J^{abs}.

Protocol reserves are finite non-redeemable equity accounts. They may receive seed capital, fee allocations, or slashed bonds and absorb loss according to the waterfall; they are not another ordinary LP tranche.

Smart contracts operate in integer base units. The reference policy rounds provider minting and withdrawal payments down, debt-share issuance up, and sends allocation remainders to an explicit dust account. Virtual assets/shares protect the initial exchange rate from donation or inflation attacks. The minimum deposit is a maintained admission guard rather than a fixed lifetime constant: a deposit is rejected whenever the integer mint formula would issue zero shares. Equivalently, at state (V, Z) the admitted deposit must satisfy the current one-share threshold, including virtual offsets. With virtual assets v_A and virtual shares v_Z, the reference integer mint is

$$m(a; V, Z) = \left\lfloor a \frac{Z + v_Z}{V + v_A} \right\rfloor .$$

Admission requires $m(a; V, Z) \ge 1$ at the current state; otherwise the deposit reverts.

Theorem 4.4 (Integer accounting closure)**.** *For every admitted transition, the integer ledger preserves assets equal to claims up to an explicit bounded dust account. The cumulative rounding residual is deterministic and cannot be used to mint unbacked provider claims or reduce debt below the amount settled.*

5 Rates, providers, and execution commitments

Utilization is $U = B/S$, with borrowed stablecoin B and supplied liquidity S. A reference kinked rate is

$$r_b(U) = \begin{cases} r_0 + s_1 U/U^*, & U \le U^*, \\ r_0 + s_1 + s_2(U - U^*)/(1 - U^*), & U > U^*. \end{cases} \tag{11}$$

The Senior supply rate is approximately utilization times borrow rate after reserve, protocol, and LBP allocations. Accrued provider income is a receivable; it becomes withdrawable cash only when repayment settles.

Theorem 5.1 (Rationed utilization equilibrium)**.** *Let $\Phi(U)$ be the raw demand-to-supply utilization map on $[0,\overline{U}]$ and define the rationed self-map*

$$\widetilde{\Phi}(U) = \min\{\overline{U}, \max\{0, \Phi(U)\}\}.$$

If demand and supply are continuous, $\widetilde{\Phi}$ has a fixed point. Any fixed point strictly below $\overline{U}$ is also a fixed point of the raw map. If $\widetilde{\Phi}$ is a contraction, the rationed fixed point is unique and globally stable; existence for the raw map additionally requires that Φ map the admissible interval into itself.

Proof. Continuity of demand and supply makes $\widetilde{\Phi}$ a continuous self-map of the compact interval $[0,\overline{U}]$, so a fixed point exists. If the fixed point lies below the clipping boundary, clipping is inactive and it solves the raw equation. The contraction claim follows from Banach's theorem. □

Utilization alone is not a complete risk price. During stress, demand can fall while expected loss rises, producing a lower utilization rate exactly when a risk spread should increase. A deployment therefore needs a risk-sensitive component based on market class, leverage, concentration, close horizon, and settlement quality.

LBP participation depends on enhanced compensation exceeding expected impairment, lock-up, and outside opportunity. The LBP is the first external provider-capital layer ahead of Senior principal, not "insurance."

A market maker may post a collateralized executable commitment specifying a quantity-price curve, validity interval, and slashing rule. Such capacity can reduce reserve demand only when delivery is enforceable and not double counted with public book depth.

Proposition 5.2 (Capacity-induced shortfall)**.** *Additional market-maker capacity can increase aggregate loss when it expands admitted credit faster than it reduces loss per admitted unit. Backstop capacity must therefore be coupled to an admission-elasticity limit rather than credited one-for-one as reserve substitution.*

6 Loss allocation, isolation, and withdrawals

Let realized shortfall after position assets be $\ell \geq 0$. The protocol applies an ordered waterfall to trader/position buffers, segregated reserves, LBP capital, global reserves when configured, and finally Senior principal.

Definition 6.1 (Ordered waterfall)**.** *For eligible allocable capacities $c_1, \ldots, c_m \geq 0$ remaining after encumbrances, reservations, required buffers, and policy haircuts, define recursively*

$$a_k = \min\left\{c_k, \left[\ell - \sum_{j<k} a_j\right]^+\right\}. \tag{12}$$

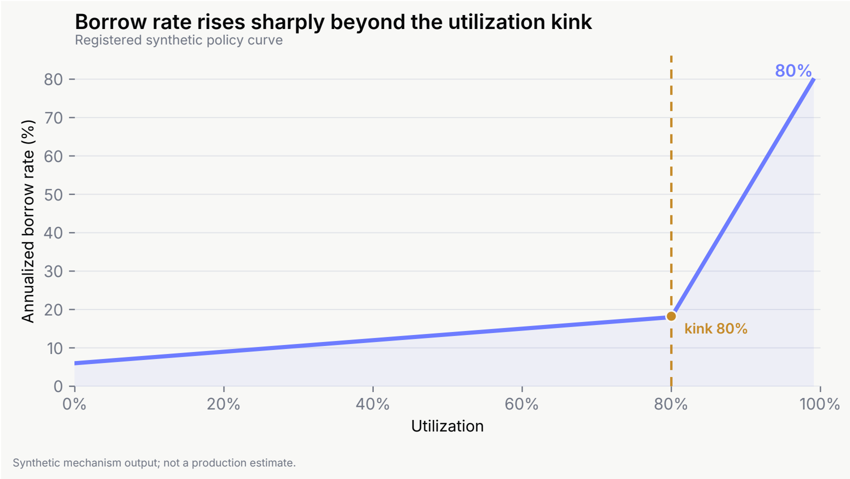

Figure 1: Illustrative utilization-based rate curve. A separate risk spread is required when expected loss and utilization move in opposite directions.

The residual is $\ell - \sum_k a_k$.

Theorem 6.2 (Existence, uniqueness, and conservation)**.** *The ordered waterfall has a unique allocation. Each layer absorbs no more than its capacity, no later layer absorbs loss while an earlier layer has unused capacity, and total absorbed loss plus residual equals* ℓ.

Corollary 6.3 (Junior-before-Senior impairment)**.** *If LBP capital precedes Senior principal, no Senior loss is recognized while unexhausted eligible LBP capacity remains.*

Strict pool isolation means one pool's local debt and provider claims cannot become another pool's liabilities. Isolation prevents direct ledger contagion but not shared shocks: two pools using the same stablecoin, venue, oracle, or market-maker sector can lose together. A shared global reserve creates protection contagion because one pool's loss can consume capacity otherwise available to another.

Let G_t be the unencumbered protocol-global reserve and let $g_{p,t}$ be the amount reserved to pool p under a frozen allocation ledger. An admissible allocation satisfies

$$g_{p,t} \geq 0, \qquad \sum_p g_{p,t} \leq G_t. \tag{13}$$

The same capital unit cannot simultaneously enter $g_{p,t}$, a local pool reserve, or collateral securing an execution commitment.

Criterion 6.4 (No protection reuse)**.** *A unit of shared protection or junior provider capital may support at most one live certificate before it is atomically released or consumed. Any pool certificate that counts an unreserved portion of* G_t*, counts the same unit in two protection layers, or treats a unit reserved in* J^{abs} *as simultaneously available in* J^{free}*, is invalid.*

For pool p at scenario start t_0, let $R^M_{p,0}$ denote eligible market/event/venue reserve capacity frozen for that pool, $J^{\text{abs}}_{p,0}$ the junior loss-absorbing capacity from definition 4.3, $R^P_{p,0}$ eligible pool-reserve capacity, and $g_{p,0}$ the protocol-global reserve allocation recorded under (13). None of these capacities includes the same capital unit twice. Junior value counted in $J^{\text{abs}}_{p,0}$ cannot simultaneously be withdrawal-service headroom, collateral for another certificate, or another protection layer.

For a scenario ω, let $\ell^{\text{prot}}_{p,j}(\omega)$ be the jth recognized shortfall reaching protocol capital after position assets and enforceable recoveries. Define cumulative recognized protocol-capital loss

$$\mathsf{CumLoss}_p(\omega) = \sum_j \ell^{\text{prot}}_{p,j}(\omega). \tag{14}$$

All eligible capacities in the following result are frozen at scenario start; replenishment, later fee income, unconfirmed receivables, completed withdrawals, and draws assigned to other pools are excluded. Queued shares included in $J^{\text{abs}}_{p,0}$ remain outstanding and loss-participating and cannot be paid and burned while the certificate remains live.

Theorem 6.5 (Scenario-conditional Senior protection)**.** *Fix a registered scenario family Ω^*_p and scenario-start capacities $R^M_{p,0}$, $J^{\text{abs}}_{p,0}$, $R^P_{p,0}$, and $g_{p,0}$. Assume the waterfall mandate makes $J^{\text{abs}}_{p,0}$ legally and technically impairable before Senior principal, the allocation satisfies criterion 6.4, and any queued shares included in $J^{\text{abs}}_{p,0}$ remain loss-participating until certificate release or impairment. If*

$$R^M_{p,0} + J^{\text{abs}}_{p,0} + R^P_{p,0} + g_{p,0} \geq \sup_{\omega \in \Omega^*_p} \mathsf{CumLoss}_p(\omega), \tag{15}$$

then Senior principal is not impaired in any scenario in Ω^*_p under the registered waterfall. The result is conditional on the stated family, mandate, queue rule, and frozen capacities; it is not a universal solvency guarantee.

Proof. By the ordered-waterfall theorem, each recognized shortfall first consumes the eligible pre-Senior layers. The junior term is the frozen impairment-eligible quantity $J^{\text{abs}}_{p,0}$, not withdrawal headroom. Because capital counted in the certificate cannot be serviced, reused, replenished, or replaced by an unconfirmed receivable during the scenario, cumulative pre-Senior capacity is the left-hand side of (15). The assumed bound therefore prevents cumulative recognized loss from reaching Senior principal. □

Remark 6.6 (Why withdrawal headroom is not protection)**.** *The protection theorem cannot use J^{free} in place of J^{abs}. For example, if other pre-Senior capacity is* 10*, $J^{\text{free}} = 80$, $J^{\text{abs}} = 20$, and cumulative loss is* 60*, the condition using J^{free} appears to pass (*$90 \geq 60$*) although only* 30 *of actual pre-Senior loss-bearing capacity exists. Senior principal is then impaired by* 30*.*

Remark 6.7 (Cumulative versus peak protection need)**.** *The theorem uses cumulative recognized protocol-capital loss because capacity is frozen and not replenished inside the*

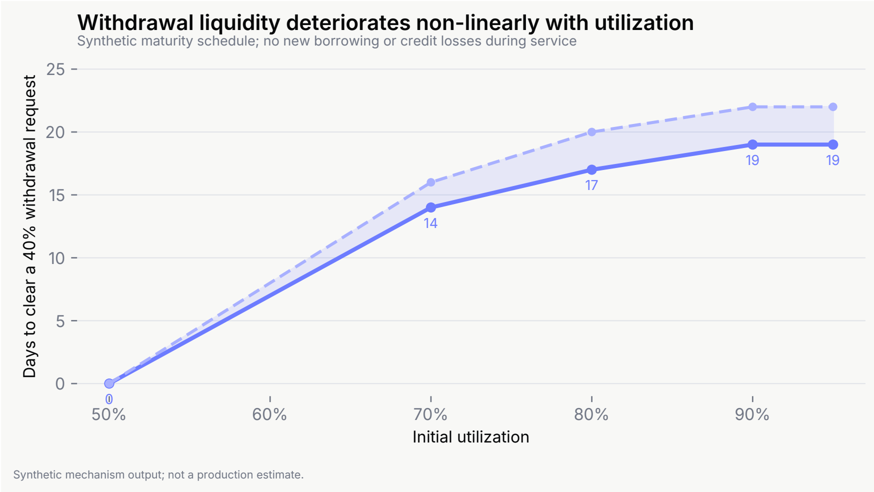

Figure 2: Illustrative loss-participating withdrawal queue. Requested shares remain exposed until service and burn.

scenario. A peak-simultaneous-loss certificate is a different object and does not imply the cumulative guarantee when capital can be consumed repeatedly.

Provider withdrawals are requested in shares, not fixed cash. Requested shares remain outstanding and loss-participating until actual service and burn. If their value is reserved in J^{abs}, service is deferred until the certificate is released or recomputed. Conservative withdrawal NAV subtracts pending-loss, liquidity, and settlement holdbacks.

Proposition 6.8 (No first-exit loss transfer)**.** *Once a risk state or holdback is publicly recognized, requesting withdrawal does not exempt queued shares from subsequent write-downs before service. Under an active protection certificate, a request alone does not reduce J^{abs}; service that would reduce that quantity requires certificate release or recomputation before payment and burn. Hence an LP cannot shift an already manifested loss to slower providers merely by entering the queue first.*

Queue clearance depends on available cash, reserve policy, and ongoing loss. A large cash buffer shortens waits but reduces lendable capital; this is a liquidity-policy trade-off, not a solvency theorem.

7 Leverage and capital efficiency

Debt fraction and gross exposure per unit of credit are

$$\delta(L) = \frac{L-1}{L}, \qquad \frac{N}{D} = \frac{L}{L-1}. \tag{16}$$

Thus a 1 unit credit pool supports 2.00 gross exposure at $2\times$, 1.50 at $3\times$, and 1.25 at $5\times$. Higher leverage is not only riskier; it consumes lender capital less efficiently.

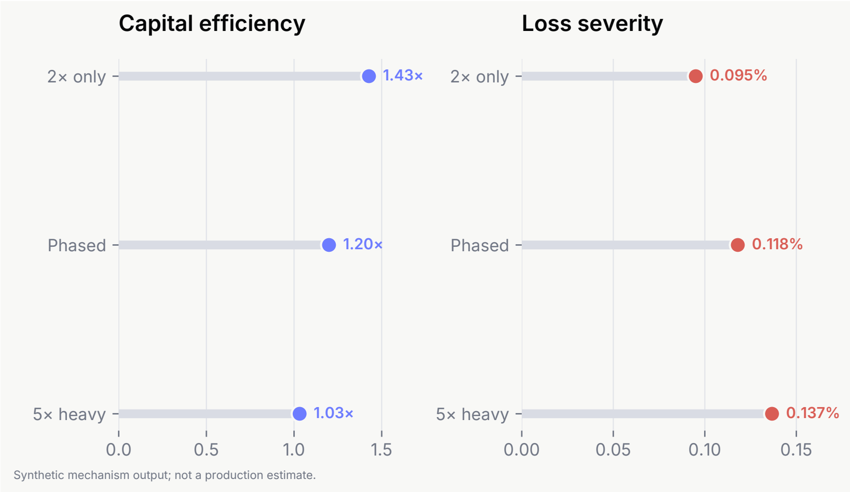

Figure 3: Illustrative capital-efficiency trade-off across leverage mixes.

For a mixed policy with gross requested exposure G_k at leverage L_k, credit demand is

$$D^{\text{req}} = \sum_k G_k \frac{L_k - 1}{L_k}. \tag{17}$$

Admission additionally requires position-level hard-flat certificates, aggregate shared-book capacity, market and account caps, reserves, and venue capabilities. The effective leverage limit is the minimum of all gates, never a universal user entitlement.

8 Endogenous capital and agent-based validation

Static accounting does not show whether providers enter, withdraw, or run. The agent model therefore endogenizes Senior and LBP supply, trader demand, market-maker delivery and strategic withdrawal, liquidator entry, reserve replenishment, and common shocks. The model contains three isolated pools (macro, election, sports), four protocol configurations, and three leverage policies. Each of 96 fixed-seed paths spans 104 weeks and contains 636 heterogeneous agents. The release records 70,207,488 scalar invariant evaluations with zero failures.

The experiment is deliberately synthetic. Behavioral rules, shock frequencies, outside options, and loss severities are author-specified. Results compare mechanisms under the same generator; they are not estimates of production probabilities or APY.

8.1 Agent state and weekly clearing

Each weekly path evolves a state vector containing pool cash, outstanding debt, Senior and LBP shares, reserves, withdrawal queues, market depth, leverage mix, market-maker commitments, and common shocks. Traders draw heterogeneous edge, size,

Table 1: Selected fixed-seed outcomes under the phased leverage mix.

Metric	Senior only	Full configuration
Two-year path Senior-loss incidence	99.0%	57.3%
Mean cumulative Senior impairment / deployed capital	40.8%	17.4%
Accepted requested credit	–	38.0%
Synthetic annualized Senior return	–	20.6%
LBP impairment incidence	–	94.8%

Table 2: Synthetic leverage-policy comparison under the full configuration. Values are properties of the registered agent model, not estimates of live market behavior.

Policy	Senior return	Senior-loss incidence	Accepted credit	LBP impairment
2× only	17.5%	22.9%	59.3%	–
Phased	20.6%	57.3%	38.0%	94.8%
5× heavy	20.9%	59.4%	34.5%	95.8%

leverage preference, and risk tolerance. Senior LPs compare expected net return with an outside opportunity and perceived loss. LBPs additionally price encumbrance and junior impairment. Market makers choose ordinary delivery and may withdraw strategically in registered states. Liquidators enter when bounty exceeds synthetic execution cost.

The weekly sequence is:

(1) update regime and common factors;
(2) update provider participation and service prior queues;
(3) generate trader demand and quote risk-sensitive rates;
(4) apply account, market, capital, and exit-capacity admission gates;
(5) realize execution, settlement, liquidation, and loss allocation;
(6) replenish reserves from designated income;
(7) update queues, provider returns, and next-period beliefs;
(8) evaluate accounting, ordering, and conservation invariants.

The model registers hypotheses before the release run. A hypothesis passes only if the stated metric crosses its pre-declared threshold. The independent daily model uses a different horizon, pool aggregation, and scenario generator, so it is a robustness implementation rather than a second sample from the same stochastic process.

Layered protection lowers Senior loss but does not eliminate it. The high synthetic return is compensation for high modeled risk, not a yield forecast. Mandate-eligible LBP capital is repeatedly impaired and replenished; withdrawal headroom is not a substitute for loss-absorbing capital, and neither can be marketed as passive insurance. Within the registered synthetic economy, neither the phased nor the 5×-heavy policy is justified by the modeled trade-off relative to 2× only: each gains roughly three percentage points of

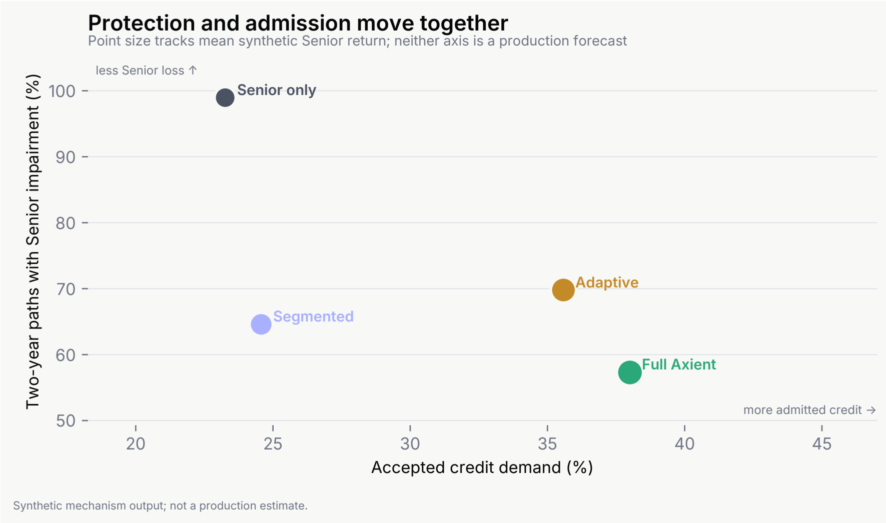

Figure 4: Synthetic protection-demand frontier. Added protection reduces Senior loss but rejects credit demand and repeatedly impairs junior capital.

Senior return while sharply increasing Senior-loss incidence, reducing accepted credit, and heavily impairing junior capital.

Six of seven registered hypotheses pass. The failed hypothesis concerns bonded market-maker capacity: modest commitment improves unit delivery but expands admitted demand enough to worsen aggregate shortfall in strategic-withdrawal weeks. The result supports the capacity-induced-shortfall proposition.

Using the displayed table values, the 5×-heavy policy raises Senior-loss incidence by approximately 159.4% relative to 2× only, while the accepted-demand ratio falls. The registered model therefore supports 2× as the only currently justified baseline; any 3× or 5× tier requires additional external evidence and a separately passing operating-region certificate.

Loss-participating queues eliminate request-time advantage in the deterministic fixture, but run-like dynamics remain when peer sensitivity is high and immediately serviceable cash is low. In stablecoin shock states, mean cross-pool raw-shortfall correlation is 0.842: local ledgers isolate obligations, not common factors.

An independently coded daily model reproduces the direction of layered protection, the leverage/hard-flat trade-off, and the withdrawal-buffer phase transition while producing materially different numerical levels. The disagreement is evidence of model risk, not a nuisance to be averaged away.

9 Contract and security boundary

The financial source of truth should contain:

(1) ‘CreditPool’ Senior shares and a loss-participating queue;

(2) 'LBPVault' junior shares, encumbrance, impairment, and queue;
(3) non-redeemable 'ReserveVault' balances;
(4) collateral-locked position accounts;
(5) debt shares and a borrow index;
(6) typed settlement evidence with replay and conflict domains;
(7) risk and venue-capability registries;
(8) liquidation, fee, and waterfall modules; and
(9) timelocked governance and emergency pause.

Backend agents may ingest books, construct quotes, route orders, monitor finality, and submit permitted transactions. They must not be able to rewrite ownership, debt, provider shares, or loss priority. External venue execution remains capability-dependent: permissionless submission at the Axient contract boundary does not imply that an operator-gated venue permits arbitrary external addresses to trade.

Principal invariants include:

- debt cannot decrease before settlement-confirmed value is admitted;
- collateral cannot escape while debt is positive;
- residual cash belongs to the trader after debt and admitted fees;
- reserves are not provider shares;
- LBP impairment precedes Senior loss;
- queued shares remain loss-participating until payment and burn;
- junior withdrawal headroom J^{free} is distinct from frozen loss-absorbing capital J^{abs};
- shared-book depth is consumed once;
- shared global protection capital is reserved once under (13); and
- integer rounding and dust are explicit and bounded.

The formal architecture does not establish deployed-contract safety. External smart-contract, economic, and operational review remain necessary.

10 Limitations and conclusion

The paper does not calibrate venue liquidity, settlement latency, market-maker behavior, provider elasticity, or loss probabilities. Synthetic agent outcomes are not commercial forecasts. Smart contracts cannot remove venue suspension, oracle error, stablecoin impairment, custody failure, legal subordination, strategic liquidity withdrawal, or common-factor shocks.

The architecture also creates trade-offs. Protection reduces admitted demand; cash buffers reduce capital efficiency; LBP compensation may be expensive; a global reserve weakens pool isolation and must be reserved without double pledge; and aggressive market-maker credits can induce more risk. Senior principal remains exposed after all prior layers are exhausted.

Table 3: Principal result and the design property it supports.

Result	Protocol implication
Settlement-confirmed debt priority	Match, attestation, or failed settlement cannot reduce debt.
Trader residual ownership	Value remaining after debt and admitted fees cannot be diverted to providers or reserve.
Integer accounting closure	Rounding is explicit, bounded, and unable to mint unbacked claims.
Unique ordered waterfall	Loss allocation is deterministic, conservative, and auditable.
Junior-before-Senior impairment	Senior protection cannot bypass unexhausted eligible LBP capital.
No first-exit loss transfer	Queued provider shares remain exposed until actual service and burn.
No protection reuse	One unit of shared reserve, commitment collateral, or junior protection capital can support at most one live certificate.
Scenario-conditional Senior protection	Frozen mandate-eligible J^{abs} and non-reused reserves protect Senior principal only within the registered scenario family.
Rationed utilization equilibrium	The clipped utilization self-map has a fixed point; a raw-map equilibrium requires the self-map condition.
Capacity-induced shortfall	Backstop capacity cannot justify unconstrained admission expansion.
Common-factor covariance	Ledger isolation does not eliminate losses from a shared stablecoin, venue, or oracle.

Within those limits, the protocol supplies a coherent answer to the funding question created by leveraged event positions. Real stablecoin credit finances real outcome exposure; on-chain priority makes provider rights inspectable; settlement-confirmed accounting prevents matched proceeds from masquerading as cash; junior capital and reserves allocate tail losses explicitly; and share-denominated queues prevent obvious first-exit transfers. The agent results reinforce a conservative product policy: start with isolated $2\times$ credit, treat $3\times$ and $5\times$ as evidence-gated tiers, and never substitute synthetic yield or backstop capacity for a robust exit and loss analysis.

11 Summary of formal guarantees

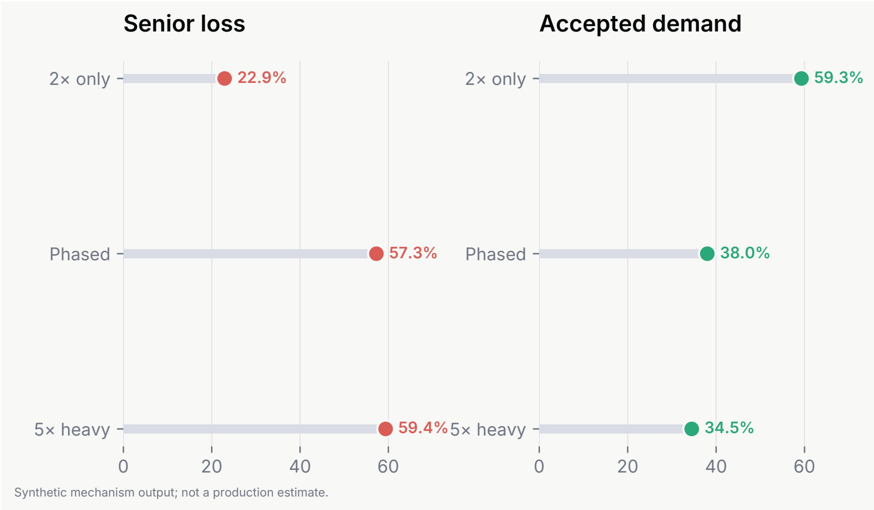

Figure 5: Synthetic leverage-policy sensitivity. Greater 5× weight raises Senior loss and reduces accepted demand.

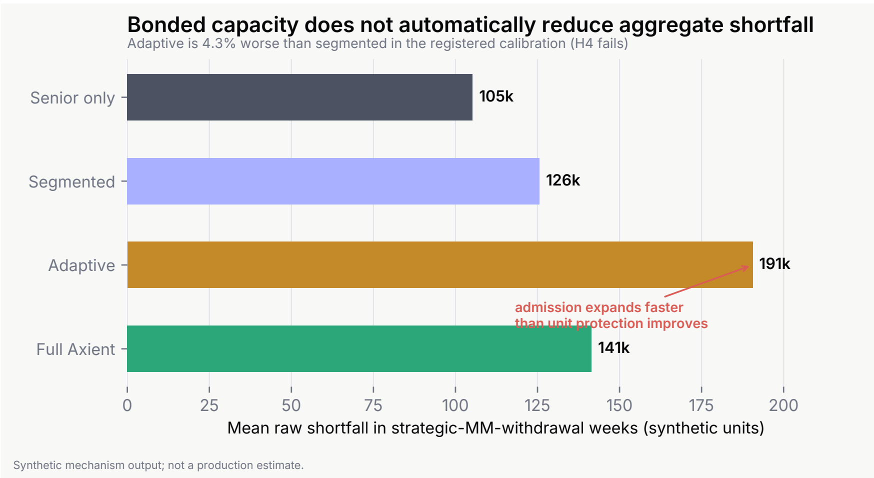

Figure 6: Registered strategic market-maker withdrawal result. Modest bonded capacity fails to reduce aggregate shortfall after admission expands.

Proof structure. The accounting results are algebraic: every admitted transition changes assets and claims by matched amounts, with residuals assigned by a fixed priority. Waterfall uniqueness follows by induction over ordered capacities. Queue neutrality follows because requested shares remain in total supply and are priced at service, not request; junior value counted in an active protection certificate remains unavailable for withdrawal service. Rationed equilibrium existence follows from continuity of the clipped self-map on a compact interval; uniqueness and global stability require contraction, while a raw-map equilibrium additionally requires the un-clipped map to remain inside the admissible interval. Contagion results distinguish direct ledger identity from covariance induced by common factors or a shared reserve.

Data and code availability

No live Axient or venue data are used. The deterministic fixtures, synthetic agent outputs, and reproducibility materials, including the $J^{\mathrm{abs}}/J^{\mathrm{free}}$ checks, are included with this revision and described in the extended arXiv version history. Archival DOI information will be added in a later revision.

Disclosure

AXIENT is an active research-and-development initiative of the author. Automated language and code-assistance tools were used for editorial review, consistency checks, LaTeX support, simulation development, and deterministic verification. The author specified and reviewed the model, assumptions, proofs, simulation rules, interpretation, and final manuscript, and takes responsibility for the work.

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